MTH 202 Chapter 3: Partial Derivatives

3.2 Functions of Two or More Variables Worksheet

1. Let $f(x,y)=\dfrac{xy^2}{3x^2+y^4}$. Determine the limits in the problem below. Make sure you write a full argument to justify your position.

a) Determine $\lim\limits_{y\to 0} f(y^2,y)$.

b) Determine $\lim\limits_{(x,y)\to(0,0)} f(x,y)$.

c) Determine $\lim\limits_{(x,y)\to(1,0)} f(x,y)$

2. Determine whether each of the following limits exists. Make sure you write a full argument to justify your position.

a) $\lim_{(x,y)\to(0,0)} \dfrac{5x^2y^2}{x^4+y^4}$.

b) $\lim_{(x,y)\to(0,0)} \dfrac{\sin\left(\sqrt{x^2+y^2}\right)}{\sqrt{x^2+y^2}}$

c) $\lim_{(x,y)\to(0,0)} \dfrac{y^4\cos^2(x)}{x^4+y^4}$

d) $\lim_{(x,y)\to(0,0)} \frac{xy}{\left(x^2+y^2\right)^2}$

e) $\lim_{(x,y)\to(0,0)} \frac{xy-x^2y}{x^2+y^2}$

f) $\lim_{(x,y)\to(1,1)} \frac{y^2}{x^2+y^2}$

g) $\lim_{(x,y)\to(1,1)} \dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$.

3. Consider the following piecewise functions $f(x,y)$ and determine $\lim_{(x,y)\to(0,0)} f(x,y)$.

a) $f(x,y)=\begin{cases} \dfrac{xy}{x^2+y^2} & \text{if} \quad y\neq x^2 \\ \dfrac{x^3}{x^2+1} & \text{if} \quad y=x^2 \end{cases}$

b) $f(x,y)=\begin{cases} \dfrac{x^2y^2}{x^2y^2+(x-y)^2} & if \quad y\neq x \\ \dfrac{x^2}{x^2+1} & if \quad y=x \end{cases}$ *Hint: Consider the path* $y=x+x^2$ *at some point.*

c) $f(x,y)=\begin{cases} \dfrac{x^2y^2+1}{x^3+y^3+1} & if \quad x^3+y^3\neq 0 \\ x^2-xy & if \quad x^3+y^3=0 \end{cases}$

d) $f(x,y)=\begin{cases} \dfrac{x^3+y^3}{x^2+y^2} & if \quad y \neq x^3 \\ \dfrac{2x^2y+1}{x^4+y^2} & if \quad y = x^3 \end{cases}$

e) $f(x,y)=\begin{cases} \dfrac{x^2+y^2}{x^2+xy+y^2} & if \quad y \neq -x \\ \dfrac{x^2-xy+y^2}{x^2+y^2+1} & if \quad y = x \end{cases}$

4. Let F be a function of x and y. Consider the following statements.

   A. $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along every straight line through $(0,0)$, but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

   B. $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along the lines $x=0$ and $y=0$, but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

   C. $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along the lines $x=0$ and $y=0$, but $\lim_{(x,y)\to(0,0)} f(x,y) = 1$.

   Which statements could *possibly* be true?

5. Mark exactly one box corresponding to the correct ending of the sentence.

$$\text{“The limit } \lim_{(x,y)\to(0,0)} \frac{x^2y^2}{x^4+y^4} \ \cdots$$

- ☐ ... does not exist because the limits as one approaches $(0,0)$ along the lines $x=0$ and $y=x$ are different.”
- ☐ ... does not exist because the limits as one approaches $(0,0)$ along the curves $y=x^2$ and $x=y^2$ are different.”
- ☐ ... exists because $\frac{x^2y^2}{x^4+y^4}$ is a composition of continuous functions”
- ☐ ... exists because the limits as one approaches $(0,0)$ along the lines $y=x$ and $y=-x$ are the same.”

6. Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$. Suppose that $f(x,y) \to 3$ as $(x,y) \to (0,1)$ along every line of the form $y = kx + 1$. What can you say about the limit $\lim_{(x,y)\to(0,1)} f(x,y)$? Select all that apply.

- [ ] It exists and is equal to 3.
- [ ] We cannot determine if the limit exists, but if it does, the limit is 3.
- [ ] It does not exist.

7. Determine 5 different paths along which you can compute the limit $\lim_{(x,y)\to(2,3)} f(x,y)$.

8. Calculate the following limit $\lim_{(x,y)\to(0,0)} \dfrac{e^{-1/\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}$.

9. Is the function $f:\mathbb{R}^2 \to \mathbb{R}$ continuous at $(0,0)$?

a) $f(x,y)=\begin{cases} x^2+y+1 & \text{if } (x,y)\neq(0,0) \\ 0 & \text{if } (x,y)=(0,0) \end{cases}$

b) $f(x,y)=\begin{cases} \dfrac{x^2y^3}{2x^2+y^2} & \text{if } (x,y)\neq(0,0) \\ 1 & \text{if } (x,y)=(0,0) \end{cases}$

c) $f(x,y)=\begin{cases} \dfrac{xy}{x^2+xy+y^2} & \text{if } (x,y)\neq(0,0) \\ 0 & \text{if } (x,y)=(0,0) \end{cases}$